

**VIETNAM NATIONAL UNIVERSITY – HO CHI MINH CITY
INTERNATIONAL UNIVERSITY
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING**



**WEB APPLICATIONS DEVELOPMENT
IT093IU
LAB 8**

Member List:

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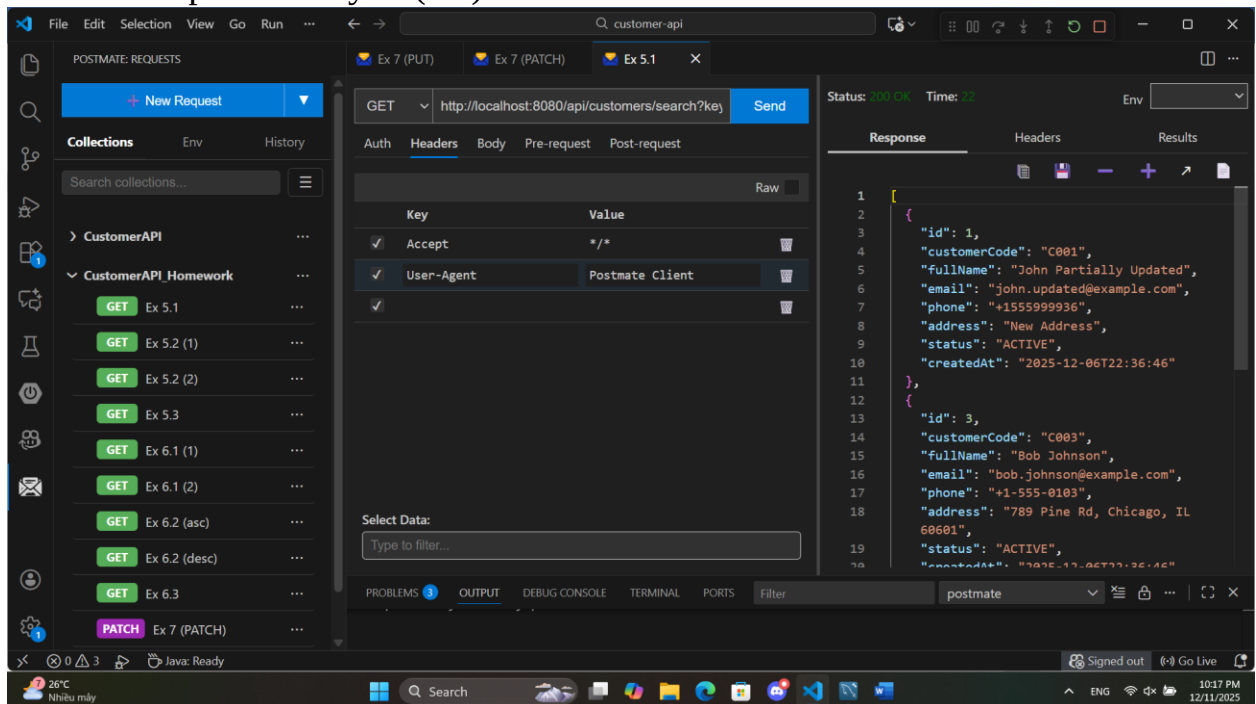
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Exercise 5

1. Task 5.1

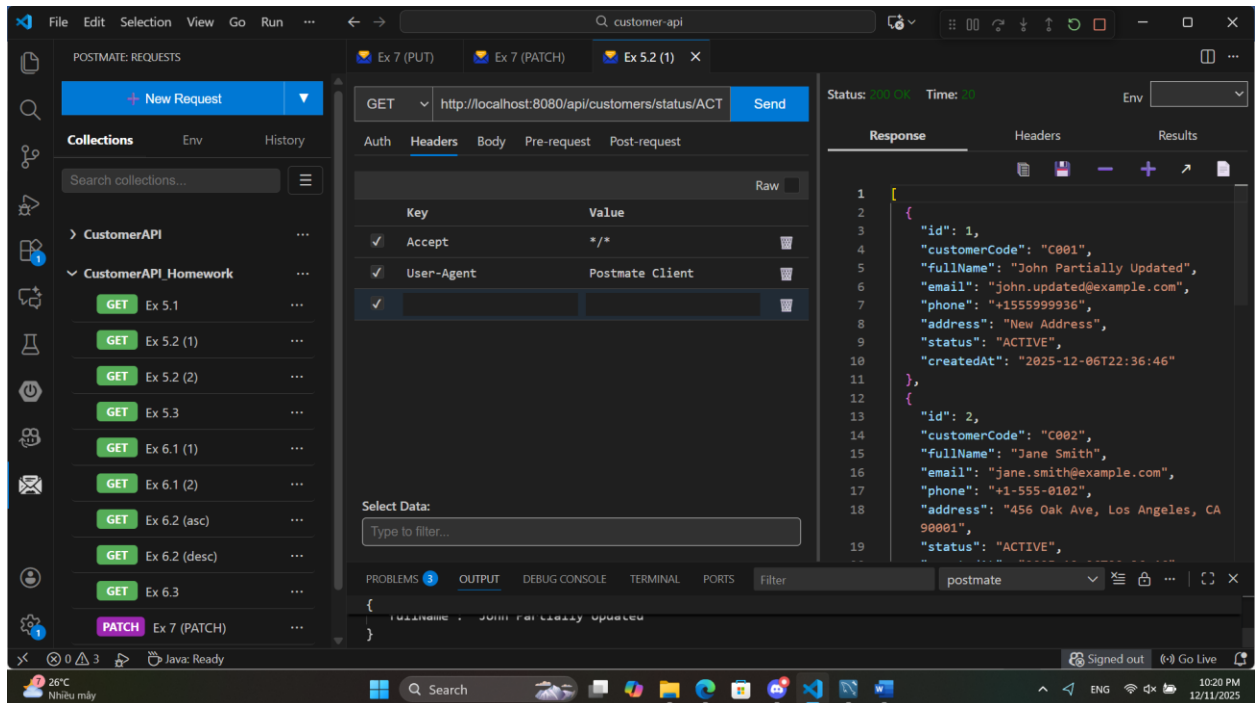
- Controller receives request at: `@GetMapping("/search")`
- Calls service:
`customerService.searchCustomers(keyword);`
- Service runs repository search query:
`SELECT c FROM Customer c`
`WHERE LOWER(c.fullName) LIKE %keyword%`
`OR LOWER(c.email) LIKE %keyword%`
`OR LOWER(c.customerCode) LIKE %keyword%`
- Entity results are converted to DTOs using `convertToResponseDTO()`.
- Returns `ResponseEntity.ok(list)`.



2. Task 5.2

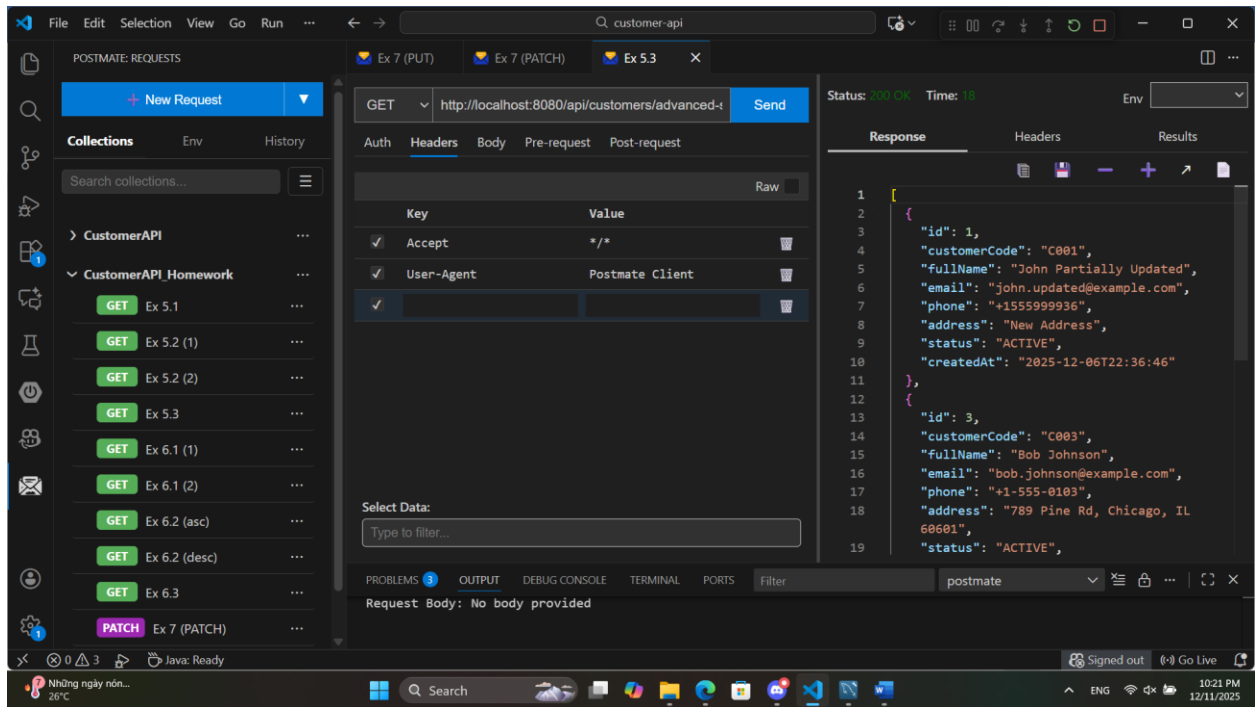
- Controller handles: `@GetMapping("/status/{status}")`
- Service converts incoming String → Enum:
`CustomerStatus enumStatus =`
`CustomerStatus.valueOf(status.toUpperCase());`
- Repository executes:
`findByStatus(enumStatus);`
Converts results to DTOs and returns as JSON.
- Error Handling:

- Invalid status (e.g., /status/ACT) throws `ResourceNotFoundException`.
- Global exception handler returns: `{ "status": 404, "error": "Not Found" }`



3. Task 5.3

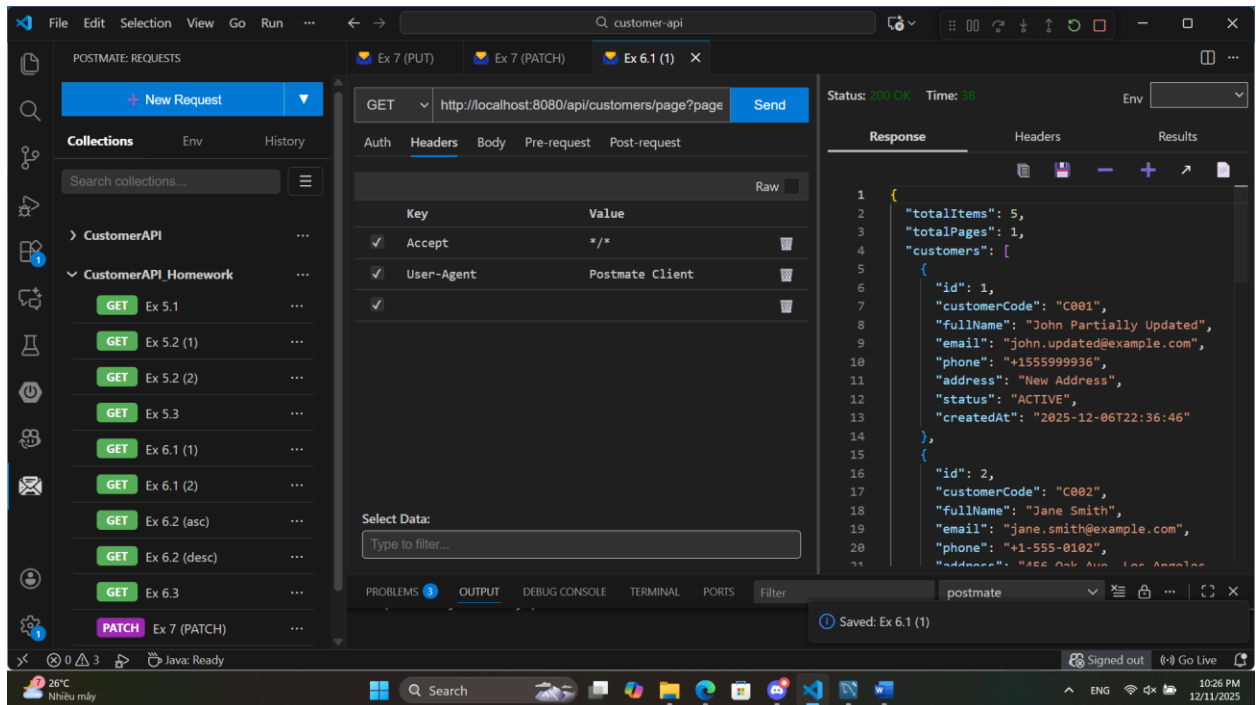
- Controller:
 - `@GetMapping("/advanced-search")`
- All fields are optional, meaning:
 - if name is provided → filter by name
 - if email is provided → filter by email
 - if status is provided → convert to enum and filter
- Service applies conditional filtering logic.
- Returns a filtered list of DTOs.



Exercise 6

1. Task 6.1

- Controller: `@GetMapping("/page")`
- Builds pageable: `PageRequest.of(page, size)`
- Repository runs: `findAll(pageable);`
- Converts each entity → DTO using mapper.
- Wraps results into response map.



2. Task 6.2

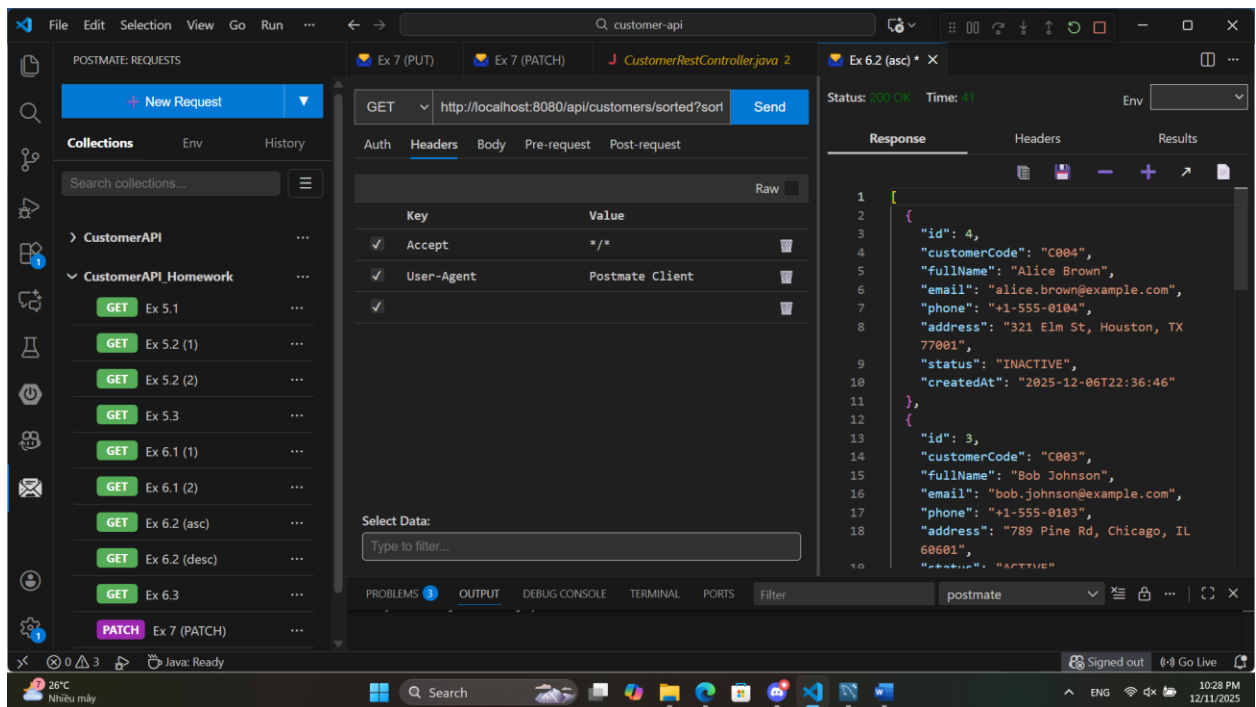
- Controller: @GetMapping("/sorted")
- Builds Sort object: Sort sort = Sort.by(sortBy).descending();
- Calls repository: findAll(sort);
- Maps entities → DTOs.
- Returns sorted list.

Test URL: /api/customers/sorted?sortBy=fullName&sortDir=asc

Returned json:

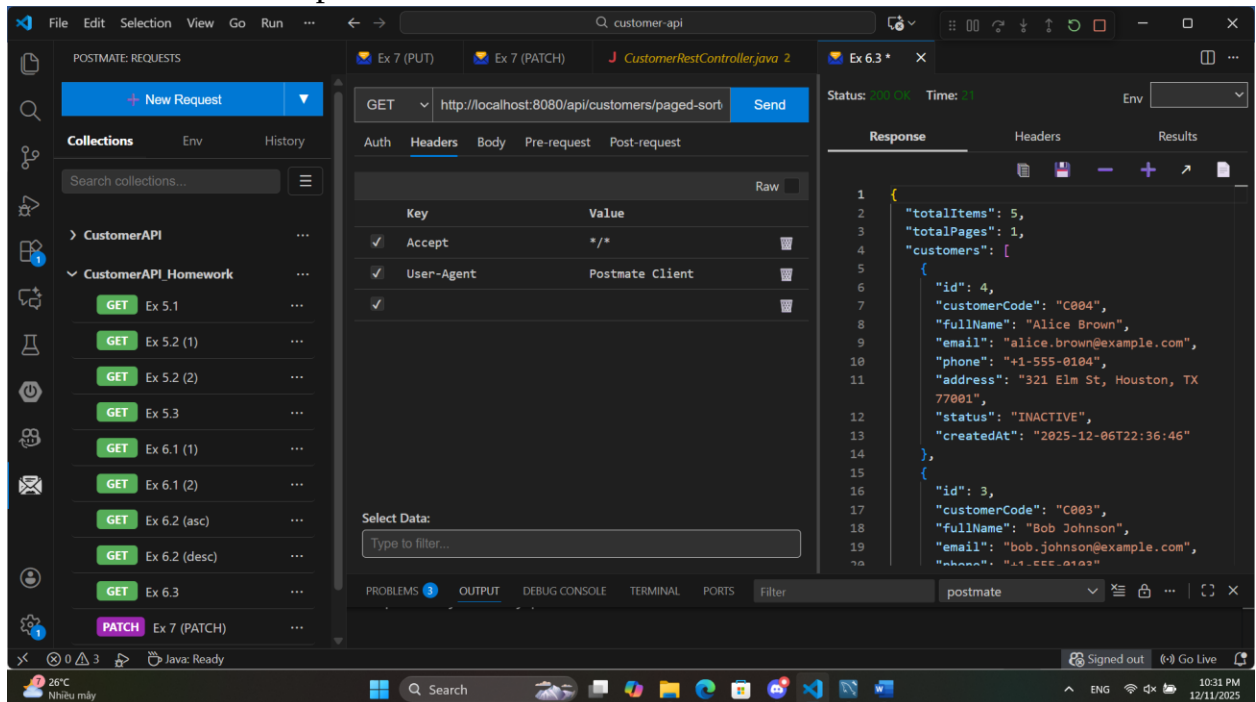
```
[
  {
    "id": 4,
    "customerCode": "C004",
    "fullName": "Alice Brown",
    "email": "alice.brown@example.com",
    "phone": "+1-555-0104",
    "address": "321 Elm St, Houston, TX 77001",
    "status": "INACTIVE",
    "createdAt": "2025-12-06T22:36:46"
  },
  {
    "id": 3,
    "customerCode": "C003",
    "fullName": "Bob Johnson",
    "email": "bob.johnson@example.com",
```

```
"phone": "+1-555-0103",
"address": "789 Pine Rd, Chicago, IL 60601",
"status": "ACTIVE",
"createdAt": "2025-12-06T22:36:46"
},
{
  "id": 5,
  "customerCode": "C005",
  "fullName": "Charlie Wilson",
  "email": "charlie.wilson@example.com",
  "phone": "+1-555-0105",
  "address": "654 Maple Dr, Phoenix, AZ 85001",
  "status": "ACTIVE",
  "createdAt": "2025-12-06T22:36:46"
},
{
  "id": 2,
  "customerCode": "C002",
  "fullName": "Jane Smith",
  "email": "jane.smith@example.com",
  "phone": "+1-555-0102",
  "address": "456 Oak Ave, Los Angeles, CA 90001",
  "status": "ACTIVE",
  "createdAt": "2025-12-06T22:36:46"
},
{
  "id": 1,
  "customerCode": "C001",
  "fullName": "John Partially Updated",
  "email": "john.updated@example.com",
  "phone": "+1555999936",
  "address": "New Address",
  "status": "ACTIVE",
  "createdAt": "2025-12-06T22:36:46"
}
]
```



3. Task 6.3

- Controller: @GetMapping("/paged-sorted")
- Combines: PageRequest.of(page, size, sort)
- Repository: findAll(pageable)
- Converts each entity → DTO.
- Builds structured response.

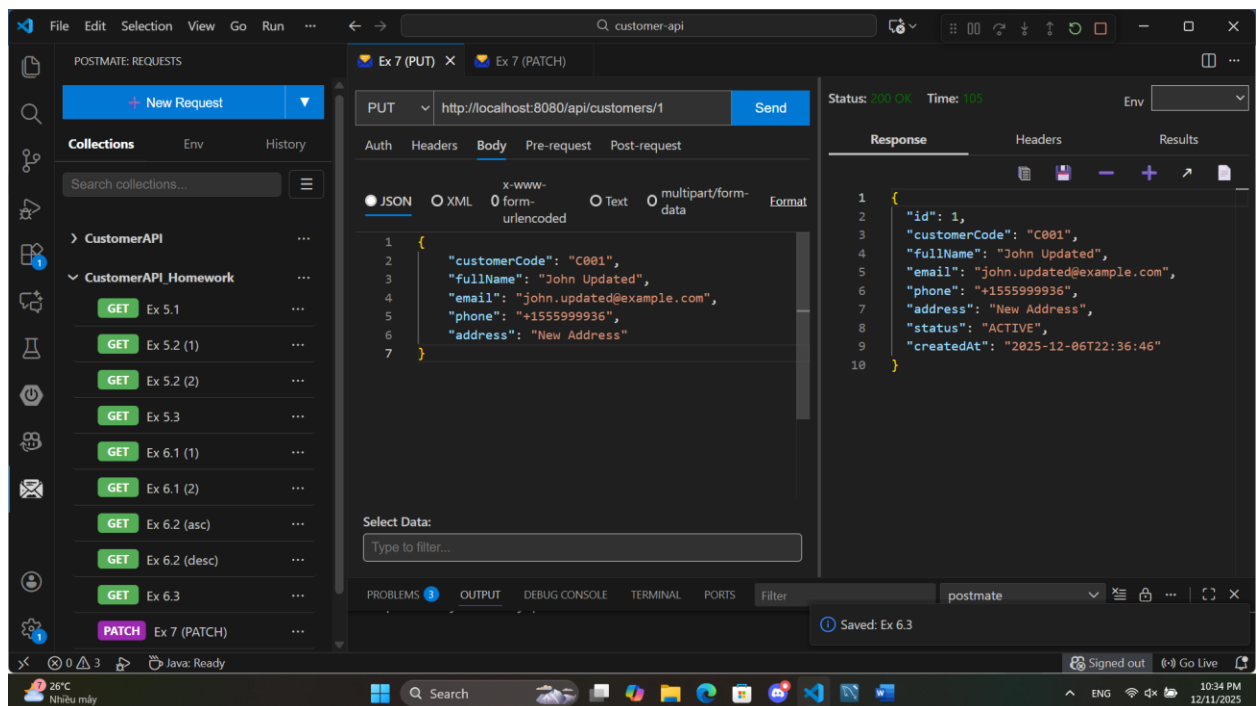


Exercise 7

- Controller: `@PatchMapping("/{id}")`
- Service loads existing customer.
- Only non-null fields in `CustomerUpdateDTO` are applied:

if (updateDTO.getFullName() != null) customer.setFullName(...)

- Email validation is performed if updated.
- Saves updated entity.
- Returns updated DTO.
- Testing: PUT vs PATCH
 - PUT
 - PUT requires complete `CustomerRequestDTO`.
 - Missing fields → validation error.
 - All fields replaced



- PATCH
 - Only `fullName` is updated.
 - Other fields remain unchanged.

